

+ Kubernetes *

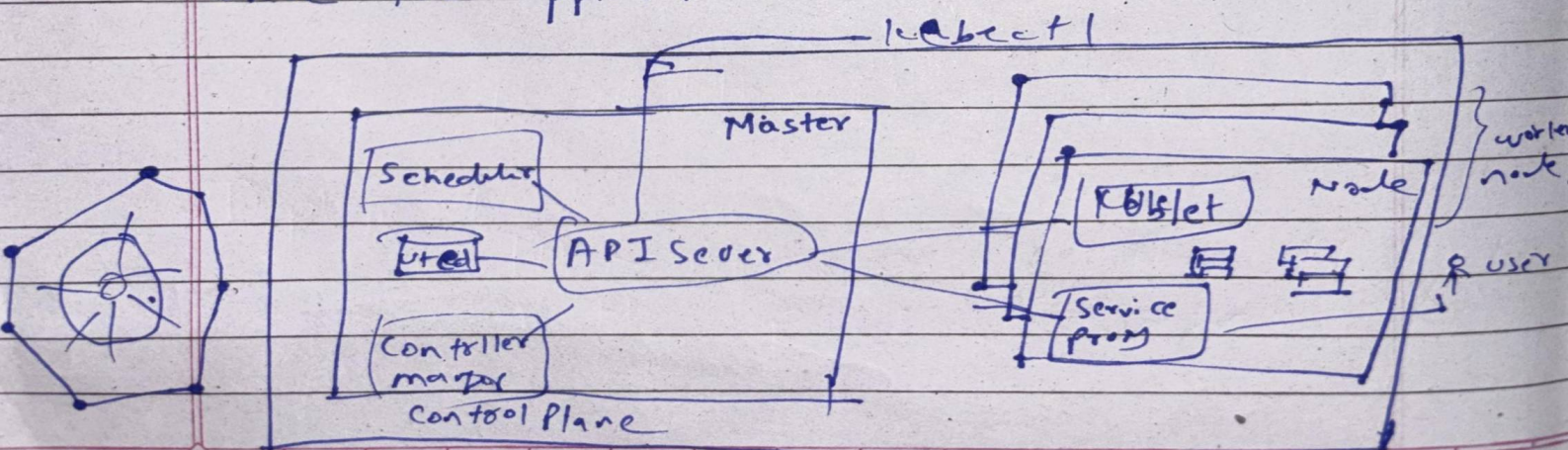
- Kubernetes is a container orchestration platform that manages the containerized microservices by automating deployment, Scaling, load balancing, & self-healing to ensure high availability.

- Microservices → Inside Containers

- Containers → managed by Kubernetes

- Kubernetes → handles Orchestration

- A Kubernetes cluster in a System made up of multiple nodes. Each node runs pods, and each pod contains one or more containers where the application runs.



Page No.: B. Ramesh
Everything in Kubernetes is manifesting file.

4

Hierarchy.

Cluster - Nodes - Pods - Containers.

Cluster:-

A Kubernetes cluster is a system of nodes where the control plane manages the workloads & worker nodes runs containers where application run smoothly.

Commands.

- ~~kubectl~~ kubectl get namespace or kubectl get ns.
- ~~set~~ kubectl apply -f <Any file yml>
- kubectl get ns or namespace
- kubectl exec -it <podname> -n <file> -- bash
- kubectl get pods -n <file>
- kubectl get pods -n <file> -o wide
- kubectl ~~get~~ scale deployment/nginx-deployment
- n nginx --replicas = 1 or 5 (desired)
- cp *.yaml to *.yml
- kubectl set image deployment/nginx-deployment
- n nginx nginx=nginx:1.27.3
- kubectl logs pod/demo-job-f -- > ^ nginx
- kubectl port-forward Service/<name> -n <namespace>
- 8080:8000 <any port> -- address = 0.0.0.0

- ReplicaSet:-

Ensures a fixed number of pods replicas are running, if a pod crashes, it creates a new one. Usually managed by a deployment.

- Scaling:-

Horizontal Scaling, increase number of pods

eg:- 20 $\rightarrow$ 50 replicas.

Vertical Scaling

increase CPU or memory of a pod.

Automatic Scaling uses:

Horizontal Pod Autoscaler (HPA).

- Stateful Set:-

It is kubernetes controller to manage stateful application that require stable identity & persistent storage.

eg:- Uses for database MySQL, PostgreSQL, MongoDB.

- Deployment:-

Manages the pods & keep the application running with the specified number of replicas.

- Rolling Deployment :-

Is a way of update on application without downtime

* In Short.

- Replica Set :-

How it works design it how much you want you can replica that much time.

or

- Replica ka kaam ye hota hai ap(user) jitna replica bologe utna vo replicas banog dega.

- Daemon Set :-

In this Set Daemon works like on every node atleast '1' replica or pod must run.

- Stateful Set :-

In this set how much replicas are present it maintains the sets with it & run it smoothly.

- Deployment :-

In this it will Rolling update, Auto-healy, Scaling etc. for the nodes.

* - The DaemonSet, ReplicaSet, StatefulSet, Deployment this all work as a same and ensures the replication controller.

- Jobs:-

A Job in Kubernetes runs a pod to do a task, finishes it, and then exits, if the pod fails Kubernetes restarts it until the task is done.

- Cron Jobs:-

A Job in Kubernetes is basically a Job that runs on schedule.

• Job - runs once, finishes, exit.

• Cron Job - Creates a job automatically at a specific time & interval.

Storage

- Persistence:-

It means data survives even if a pod restarts, crash etc.

- Persistent Volume (P.V):-

PV is actual storage in cluster.

- Persistent Volume Claim (P.V.C) -

P.V.C is a request for storage made by user & pod.

P.V.C acts like a binding layer between Pod & P.V.

- flow :-

- PV = actual storage
- PVC = request + binding
- Pod = uses PVC to write data.

- Second Scenario of flow :-

Pod $\rightarrow$ PVC $\rightarrow$ PV $\rightarrow$ Actual Storage.

Service :-

A Service in Kubernetes is a stable way to access a group of pods. It gives pods a fixed IP and DNS name and forwards traffic to them.

Use cases :-

- Exposing a web app to users.
- connecting frontend to backend
- Load balancing between multiple pods & replicas
- Allowing internal communication between microservices.

- Ingress :- Reroute the traffic in Kubernetes.

= ~~Set~~^{set} use use when the data is important

* # In most Kubernetes YAML files, the two main sections are

1) Metadata

2) Spec

These both different Role

- Metadata, is basic information about the object

eg:- name, labels, namespace

- It helps Kubernetes identify the Object

= Spec (Specification)

- The important part, it defines what you want Kubernetes create and how it should behave.

eg:-

- In Deployment:- Spec defines, replicas, Selector, template

- In Pod:- Spec define Containers

- In Service:- Spec defines Type, ports, selectors

- In Simple Understanding

metadata → who it is

Spec → what it should be

- Namespace :-

Namespace is used to Separate resources inside the Same Cluster.

- Flow of YAML

YAML

- metadata

- Spec

 - Selector

 - template

 - metadata (labels)

 - Spec

 - containers

 - ports

 - volumes

 - images

- In Spec there will be Selector & inside Selector we don't define ~~templates~~ templates we define it outside & in templates we define labels

Nginx:-

It is an open source web-server software where it serves the website and acts as a reverse proxy & distributes the traffic across multiple backend servers.

The nginx becomes powerful when instead of users directly accessing your backend app they hit nginx first.

So it acts like a middle layer between users and application.

Scaling in Kubernetes:-

It means increasing & decreasing number of pods & replicas to handle the load of application.

Type:-

i) Horizontal Pod Autoscaling:-

Increase/decrease number of pods & replicas.

ii) Vertical Pod Autoscaling:-

Increase or decrease CPU or memory of pod.

Use of case:-

If traffic increase on your e-commerce app during a sale, your scale from 2 pods to 10 pods to handle more users

Scheduling in kubernetes:-

It means or is the process where kubernetes decide that on which node a pod should run based on resource availability & req.

Use case:-

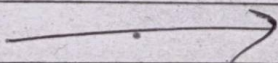
If Pod requires 2 GPU & 4GB RAM then Scheduler place that on a Node that has enough available resources.

RBAC (Role Based Access Control)

RBAC is a kubernetes authorization method that allows you to define permissions for user, groups or service account based on roles

RBAC decides who can do what inside kubernetes the cluster

Example:



Suppose you have four

- DevOps Engineer → full cluster access
 - Developer → can only deploy applications
 - Viewer → can only see resources
- RBAC lets define these permissions.

* Monolith :-

Is one big application where all components work together in a single codebase are deployed as one unit.

* Microservices :-

Break the application into small independent services, each handling a specific function & deployed separately.

Helm :-

Is an external package manager for K8s that helps define, install, and ~~manage~~ manage applications using reusable templates called charts, reducing the complexity of handling multiple YAML files

Service Mesh :-

- Controls how Services talk to each other in a Kubernetes environment.

OR

- Service mesh adds control, security and visibility to Service-to-Service communication.